

A Derivation and Analysis of the Transmission Line Speaker Enclosure

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INTRODUCTION

The problem of low frequency reproduction is an area of great importance in loudspeaker research and design. The moving coil speaker is dipolar and creates opposing variations in pressure on the two sides of the cone. If a speaker (hereafter referred to as a driver or woofer) is mounted in free air, the opposing pressures on each side of the cone will cancel, rendering it ineffective at wavelengths which are larger than its dimensions. Over the years, four major approaches to dealing with this problem have been developed, all of which involve mounting the speaker in some kind of enclosure. The enclosure types associated with these four approaches are the Sealed Box, the Vented box, the Horn, and the Transmission Line. All speaker enclosures represent variations of one or more of these four types.

The sealed box [14] is the most direct approach. As the name implies, the driver is mounted in a sealed box which completely contains the rear radiation and acts as an air spring which the cone must work against. This enclosure has been thoroughly analyzed in the literature [5,12,13] and can be modeled as a second order high pass filter. Its characteristics are well known and it is widely used.

The Vented Box differs from the Closed Box in that it attempts to invert and use the driver's rear radiation to reinforce the front radiation over a band of frequencies near the low frequency limit of the system. A ducted opening in the enclosure causes it to behave as a Helmholtz resonator [11,12] and the tuning of the resonator is a function of the duct size and length, cabinet volume, and driver parameters. It too has been thoroughly treated in the literature [5,12,13] and can be modeled as a fourth order high pass filter. Though more complex than the Sealed Box, it has somewhat greater efficiency and is also in widespread use.

The Horn enclosure, which is prevalent in professional applications, dispenses with the driver's rear radiation with a small enclosure behind the cone and loads the front with a horn. The horn acts as an acoustical lever or transformer which matches the high mechanical impedance of the driver to the low radiation impedance of the air, increasing the efficiency. Although horn design is prone to surprises which the math doesn't predict, the theory is fairly well developed [7] and again, much information is available.

The Transmission Line (TL) loads the driver with a long pipe having dimensions comparable to the lowest wavelengths that the system is expected to produce. The pipe is open at the end opposite the driver and filled with sound absorbent material which is intended to dissipate the rear radiation as completely as possible [6]. A variation of the TL is the Labyrinth, which partially absorbs the rear wave, but containing less absorbent material, allows significant amounts of delayed radiation from the line exit to reinforce the direct output of the driver. The Transmission Line is the subject of this paper.

BACKGROUND

A TL enclosure ideally is infinitely long and completely absorbs the backwave from the driver. Since this is not practical, a judiciously chosen finite length is used instead, and radiation emanating from the open end reinforces the direct output of the driver. The resonant behavior of the pipe serves to reduce cone motion at strategic frequencies. The absorption in the line increases with frequency, reducing or eliminating higher order resonances and further serving to damp the driver's fundamental resonance. In practice, this is found to be quite effective and the TL has earned for itself a reputation as a "non-resonant" enclosure [6]. The distinguishing feature of the TL is that unlike closed or vented boxes, it does not attempt to augment bass response with additional resonances.

Traditionally, the line is made $1/4$ of a wavelength long at the resonant frequency of the driver, and at resonance, the steady state output at the line exit is 90 degrees out of phase with the forward radiation and slight reinforcement occurs at this frequency and in the octave above. Also, at $1/4$ wavelength, the pipe itself, due to its resonant behavior, presents a high impedance to the driver and damps cone motion near this frequency. Below resonance, the output asymptotically approaches a 180 degree phase shift at low frequencies. However, the output at the line exit is attenuated considerably and has relatively little contribution to the total sound pressure in the room [6].

It is recommended that the cross section of the line be equal to or greater than the cone area of the woofer to minimize internal pressure, at least in the first half of the line [6]. Traditionally, the ideal TL woofer has high suspension compliance, a low resonant frequency (20-40 Hz), and low Q_t (typically 0.3) [5]. It should be noted that since these guidelines originate from experiment and tradition rather than theoretical development, numerous variations of the design exist as well [4].

Transmission lines are traditionally stuffed with long fiber wool, although fiberglass, cotton, and polyester batting are also used. Organic materials are preferred by many, presumably because the physical characteristics of the individual fibers vary from one fiber to the next, and even over the length of the fiber, producing a broad statistical distribution of absorption and resonance characteristics, while synthetic fibers have uniform thickness and are more reactive and, as Bradbury shows [2], have significant impedance fluctuations at low frequencies.

The TL has several weakness which contribute to its rarity in commercial designs. Cabinet construction is more complex than that of ordinary enclosures and TL's are typically larger and less efficient than sealed or vented systems. The TL, unlike the sealed box, does not limit cone excursion at infrasonic frequencies. Another common criticism is that TL bass response is deficient, lacking the "punch" that characterizes other designs. This of course is a two sided argument and a matter of individual taste; the TL has its pros and cons just like any other enclosure. It represents a particular set of compromises which, for subjective reasons, are favored by a sizable number of enthusiasts. The greatest weakness, though, is the absence of mathematical justification for its design.

Analysis of the TL (or Labyrinth) is fundamentally different than that of the other enclosure types because lumped parameter methods are not applicable. Because the dimensions of the line are comparable to the length of the propagating waves, the analogous electrical circuit is not simply a combination of resistors, inductors and capacitors, but an

electrical transmission line, hence its name. In contrast with the other enclosure types, no explicit mathematical model for the TL exists in mainstream audio literature; information is essentially limited to general guidelines which originate from a tradition established by Bailey and others [2,5]. The trial and error (and mystique) associated with the design of a successful TL system, though typical of the audio industry, would be considered unacceptable in other areas of the engineering profession. A clearer understanding of the physical principles which govern the Transmission Line is needed and could certainly lead to better products in the marketplace. It is therefore the purpose of this paper to develop a mathematical description of the TL, from which design guidelines can be established and meaningful analysis can be made before a system is built.

DERIVATION

The objective of this derivation is to allow the designer to specify a driver and its particular Thiele/Small parameters [5] and accurately predict the relationship between input voltage and acoustic pressure when that driver is loaded with an acoustic transmission line having a given length, cross section, and fibrous material of specified density. Furthermore, the case in which the line exit is blocked will also be considered.

In order to hold the mathematics to a manageable size and to assist in further simplifying the results, the following assumptions will be made:

1. The line loading the driver is of constant cross section and uniform stuffing density.
2. The system is radiating into 2π steradians; i.e. all sources are close to a hard boundary such as a floor.
3. All examples will assume that the distance from the various sound sources and the microphone are equal and the results normalized to one meter distance.

This derivation involves a rather large number of variables. Their definitions and units (MKS) are as follows:

Bl	Product of magnetic flux density and length of wire in voice coil gap; Newtons per Ampere
c	Speed of light; 343 meters per second unless otherwise stated
Cms	Compliance of driver suspension, meters per Newton
Eg	Voltage at driver terminals, volts
fs	Resonant frequency of driver, cycles per second
j	The square root of -1
k	Propagation constant (wavenumber), inverse meters
L	Length of transmission line, meters
Le	Voice coil inductance, Henrys
Mmd	Mass of moving assembly in a driver, kilograms
p	Acoustic Pressure, Pascals
Qe	Q factor of driver due to electrical effects, dimensionless
Qm	Q factor of driver due to mechanical effects, dimensionless
Qt	Q factor considering both electrical and mechanical effects, dimensionless
Re	Electrical resistance of voice coil, ohms
Rm	Mechanical resistance of driver suspension, Newton seconds per meter
s	$j\omega$; $\omega=2\pi f$
Sd	Area of driver cone, square meters
Sl	Cross sectional area of transmission line, square meters
ua	Linear acoustic velocity, meters per second
uf	Velocity of fibrous material in line, meters per second
ud	Velocity of driver cone, meters per second
uv	Linear acoustic velocity of air at line vent
U	Acoustic volume velocity, cubic meters per second
Vas	Volume of air having acoustic compliance equivalent to that of driver cone and suspension, cubic meters
z	Specific acoustic impedance, Newton seconds per cubic meter
Z	Acoustic impedance, Newton seconds per meter ⁵
Zf	Mechanical impedance seen by front of cone, Newton seconds per meter

Z_b	Mechanical impedance seen by rear of cone, Newton seconds per meter
Z_v	Radiation impedance at line exit, Newton seconds per meter
γ	Propagation coefficient, dimensionless
Γ	Reflection coefficient, dimensionless
λ	Wavelength, meters
σ	Drag parameter, Newton seconds per meter ⁴
ρ_F	Absolute density of fibers, kilograms per cubic meter
ρ_P	Packing density of fibers, kilograms per cubic meter
ρ_o	Density of air, 1.29 kilograms per cubic meter unless otherwise specified
ω	Radian frequency, radians per second

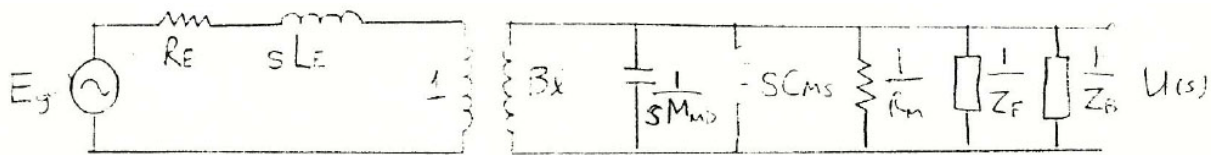
It is extremely helpful, particularly for the electrical engineer, to construct an equivalent electrical circuit representing all of the significant features of the driver. Since a driver's behavior is dependent upon the interrelation of electrical, mechanical, and acoustical elements, translation of these variables into the electrical domain is of considerable assistance in simplifying the problem.

The convention used to develop the model used in this paper is as follows:

Velocity--Voltage
 Force--Current
 Compliance--Inductance
 Mass--Capacitance
 Mechanical impedance--electrical admittance

MODELING THE DRIVER

Using the above relationships results in the following electrical circuit for a moving coil driver:



The Transfer function $u_d(s)/E_g(s)$ is

$$\frac{R_E + sL_E}{Bl} = \frac{1}{sM_{MD} + \frac{1}{sC_{MS}} + R_M + Z_F(s) + Z_b(s)} + Bl \quad (1)$$

This expression relates cone velocity to input voltage in terms of the physical properties of the driver, i.e. cone mass, suspension compliance, magnetic field strength, etc. While these terms are most useful to the driver designer, they are not as functionally descriptive as the widely used Thiele/Small parameters [5,12,13]. The following relations are helpful in translating the variables:

$$f_s = \frac{1}{2\pi\sqrt{M_{MD}C_{MS}}} \quad (2)$$

$$Q = R\sqrt{\frac{C}{L}} \quad (3)$$

$$M_{MD} = \frac{Q_E(Bl)^2}{2\pi f_s R_E} \quad (4)$$

$$C_{MS} = \frac{R_E}{2\pi f_s Q_E (bl)^2} \quad (5)$$

$$(Bl)^2 = \frac{R_E}{Q_E} \sqrt{\frac{M_{MD}}{C_{MS}}} \quad (6)$$

$$R_M = \frac{1}{Q_M} \sqrt{\frac{M_{MD}}{C_{MS}}} \quad (7)$$

$$Q_T = \frac{1}{\frac{1}{Q_E} + \frac{1}{Q_M}} \quad (8)$$

Further, a relationship between Bl and V_{as} can be determined:

$$V_{AS} = \rho_o c^2 C_{MS} S_D^2 \quad (9)$$

$$V_{AS} = \frac{\rho_o c^2 R_E S_D^2}{2\pi f_s Q_E (Bl)^2} \quad (10)$$

Substitution yields alternative expressions using the Thiele-Small parameters:

$$M_{MD} = \frac{\rho_o c^2 C_{MS} S_D^2}{4\pi^2 f_s^2 V_{AS}} \quad (11)$$

$$R_M = \frac{1}{Q_M} \frac{\rho_o c^2 S_D^2}{2\pi f_s V_{AS}} \quad (13)$$

$$C_{MS} = \frac{V_{AS}}{\rho_o c^2 S_D^2} \quad (12)$$

Z_f represents the mechanical impedance, which is imposed upon the driver by the acoustic radiation impedance of the air. In this paper, it is the ratio of force to velocity for a rigid piston [9]:

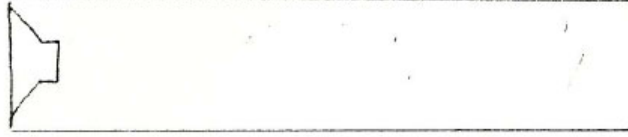
$$Z_{f(s)} = -\frac{1}{2\pi} S_D^2 \frac{\rho_o}{c_o} S^2 + \frac{8}{2\pi^{3/2}} S_D^{3/2} \rho_o S \quad (14)$$

This equation is valid for frequency below which the wavelength is relatively large compared to the dimensions of the piston.

Z_b is the impedance seen by the rear of the cone and is determined by the enclosure, and is usually the most significant term in equation (1) at low frequencies. The focus of this work is to quantify this term.

THE TRANSMISSION LINE

As stated previously, this development will only consider a straight pipe of uniform cross section and packing density:



Consider a boundary which causes a forward traveling (+x) wave, $U^+ e^{-jkx}$ to be reflected. The reflected wave, $u^+ e^{-jkx}$ propagates in the (-x) direction and is related to the incident wave by the expression

$$u^- e^{+jkx} = \Gamma(x) u^+ e^{-jkx} \quad (15) \quad \text{Or} \quad \Gamma(x) = \frac{u^-}{u^+} e^{2jkx} \quad (16)$$

where $\Gamma(x)$ is the reflection coefficient. The definition of Γ is somewhat arbitrary, as it can be defined in terms of either pressure or velocity. Since our model for the driver relates input voltage to cone velocity, it is most useful to define the reflection coefficient Γ in terms of velocity.

The boundary conditions require that at a hard boundary, for example, the acoustic velocity is zero and the reflected wave has amplitude equal to that of the incident wave. The pressure at the boundary will be maximized at a hard boundary, and the reflected pressure will be equal in amplitude to the incident pressure. Thus the reflection coefficient will be -1 if defined in terms of velocity and +1 if defined in terms of pressure.

The total velocity:

$$u(x) = u^+ e^{-jkx} + u^- e^{+jkx} = u^+ e^{-jkx} [1 + \Gamma(x)] \quad (17)$$

The total pressure:

$$p(x) = p^+ e^{+jkx} + p^- e^{-jkx} = p^+ e^{-jkx} [1 - \Gamma(x)] \quad (18)$$

And the acoustic impedance $z=p/u$ can be written as

$$z(x) = \frac{p(x)}{u(x)} = \frac{p^+ [1 - \Gamma(x)]}{u^+ [1 + \Gamma(x)]} = Z_0 \frac{1 - \Gamma(x)}{1 + \Gamma(x)} \quad (19)$$

where Z_0 is the intrinsic acoustic impedance of the medium.

The reflection coefficient $\Gamma(x)$ can now be written as

$$\Gamma(x) = \frac{Z_L - Z(x)}{Z_L + Z(x)} \quad (20)$$

at another location x' :

$$\Gamma(x') = \Gamma(x)e^{2jk(x'-x)} \quad (21)$$

A similar development for electrical transmission lines is given by Johnk [8].

The term k , in the lossless case, is $\frac{\omega}{c}$, or in the s domain may be written as $-jS/C$. (22)

An analytical expression for the impedance at the input of the line may be developed by first defining the acoustical impedances. Morse defines the characteristic acoustical impedance of a pipe as

$$Z_L = \frac{\rho_0 c}{S_L} \quad (23)$$

where S_L is the cross-sectional area of the pipe. The impedance at the mouth is equivalent to that of a massless piston of the same dimensions [3]. The mechanical impedance of a piston from eq.(14) is

$$Z_v = \frac{f}{u} \cong -\frac{1}{2\pi} S_L^2 \frac{\rho_0}{c} S^2 + \frac{8}{3\pi^{3/2}} S_L^{3/2} \rho_0 S \quad (24)$$

and the acoustic impedance is found by dividing eq.(24) by S_L^2 which converts the mechanical impedance to an acoustic impedance [9, p.231]:

$$Z_v = \frac{p}{u} \cong -\frac{1}{2\pi} \frac{\rho_0}{c} S^2 + \frac{8}{3\pi^{3/2}} S_L^{-1/2} \rho_0 S \quad (25)$$

Note that eq.(25) has the same dimensions as eq.(23). The reflection coefficient at the line exit, from (20) is

$$\Gamma = \frac{Z_L - Z_v}{Z_L + Z_v} \quad (26)$$

and the reflection coefficient at a distance L from the exit, at the driver's position in the line, from eq.(21) is

$$\Gamma(L) = \frac{Z_L - Z_v}{Z_L + Z_v} e^{2jkL} \quad (27)$$

The acoustic impedance seen by the cone, from eq.(19) is

$$Z(L) = Z_L \frac{1 - \frac{Z_L - Z_v}{Z_L + Z_v} e^{2jkL}}{1 + \frac{Z_L - Z_v}{Z_L + Z_v} e^{2jkL}} \quad (28)$$

which may be reduced to

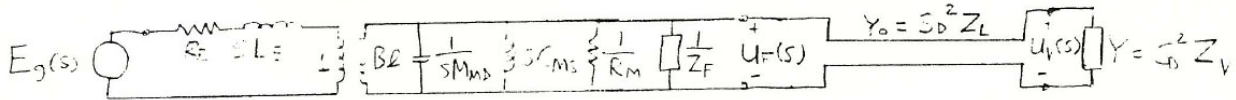
$$Z(L) = Z_L \frac{Z_v \cos(kL) + jZ_L \sin(kL)}{Z_L \cos(kL) + jZ_v \sin(kL)} \quad (29)$$

The form of this equation is identical to that given by Olson [11] for a pipe open at both ends.

Eq.(29) must be converted to a mechanical impedance if it is to be substituted into eq.(1) as Z_b . This is accomplished by multiplying eq.(29) by S_d^2 , the square of area of the driver cone, which in general is different from the cross-sectional area of the line. The mechanical impedance seen by the moving system is therefore

$$Z_B = S_d^2 Z_L \frac{Z_v \cos(kL) + jZ_L \sin(kL)}{Z_L \cos(kL) + jZ_v \sin(kL)} \quad (30)$$

It is now possible to draw a complete equivalent electrical circuit, which includes the acoustical transmission line:

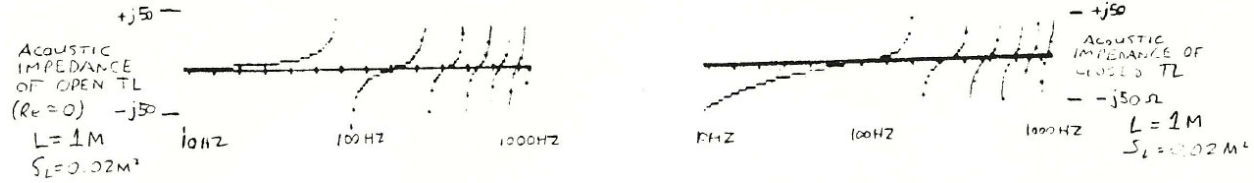


Johnk [8] gives an equation for the input impedance of an electrical transmission line which is identical to Olson's when admittances are converted to impedances. The analogy between electrical and acoustical transmission lines proves to be useful and accurate. Additional details in the analogy not considered relevant to this discussion are provided by Morse and Ingard [10].

At this point it is instructive to examine the mechanical impedance as a function of frequency for both open ended lines ($Z_v \rightarrow 0$) and close ended lines ($Z_v \rightarrow \infty$). The following are plots of eq.(29) using $S_v = 0.02 \text{ m}^2$, $L = 1 \text{ M}$, and assuming no losses. They show both real and imaginary parts, from 10Hz to 1000Hz and -50 to +50 ohms.

It is interesting (and expected) to find that at frequencies well below the first resonance of the pipe, the open ended line reduces to a mass equal to that of the air in the line, coupled to the cone. The close ended pipe presents a compliance equal to that of an acoustic suspension enclosure of the same volume. This can be verified by simplifying eq.(29) with a first order

Taylor series approximation.



The closed pipe has an impedance minima (resonance) at $f_1 = c/4L$ and harmonics at $f_2 = 3f_1$, $f_3 = 5f_1$, etc. [11] and maxima (antiresonance) at $f_1 = c/2L$ and $f_2 = 2f_1$, $f_3 = 3f_1$, etc. The open pipe has impedance maxima at $f_1 = c/2L$ and $f_2 = 2f_1$, $f_3 = 3f_1$, etc. Its impedance minima are at $f_1 = c/2L$ and $f_2 = 2f_1$, $f_3 = 4f_1$, etc.

The closed line is in many ways the inverse of the open line. The antiresonance at $\lambda/4$ in the open line has traditionally been used to limit cone motion at the woofer's resonance, but in some applications this may not be desirable; at $\lambda/4$ the closed line has a resonance. The closed line provides an alternative, which has scarcely been considered in the literature [6]. Notice that the closed line's first resonance is an octave below that of the open line. Also, notice that while the open line does not limit cone motion at infrasonic frequencies, the closed line becomes an acoustic suspension enclosure as $f \rightarrow 0$.

Now that the general nature of pipes has been discussed, it is necessary to bring fibrous materials into the discussion and examine their effects on the pipe's operation.

Bradbury's 1976 paper [2] combined the results of several earlier works with his own theory and empirical data, obtaining a model which seems to account for the main acoustical characteristics which are observed. The following development is taken from this paper [2].

The presence of fibers in a pipe creates drag such that the aerodynamic force per unit length and cross-sectional area is $\sigma(u_a - u_f)$ where u_a is the velocity of the air, u_f the velocity of the fiber, and σ is the aerodynamic drag parameter which is fixed for a particular material at a specified packing density. The equation of motion is

$$\rho_p \frac{du_f}{dt} = \sigma(u_a - u_f) \quad (31)$$

Bradbury points out that the fibrous material can be assumed to have no significant stiffness of its own. The theory is fairly complete in accounting for the combined effects of drag and the mass of the fibers on the movement of air in the line.

At extremely low frequencies, the fibers are able to move in phase with the air in the line, and dissipation of energy is minimized. The packing density of the fibers is added to the density of the air, reducing the speed of sound. In this case, the wave is not attenuated because viscous losses are minimized.

At high frequencies, the mass of the fibers prevents them from moving with the air, so drag dissipates a great deal of energy and the wave is attenuated. Since the mass of the fibers is not added to the mass of the air, the phase velocity in the fibrous tangle remains at its free-air value.

The introduction of fibrous materials to a pipe, then, changes two key parameters, the intrinsic acoustical impedance of the medium $\rho_0 c$, and the propagation constant k . Each is

multiplied by a constant which is complex and accounts both for losses at high frequencies (the imaginary part) and changes in phase velocity at low frequencies (the real part) and is given by

$$\gamma = \sqrt{\frac{1 + \rho_p / \rho_0 + S \rho_p / \sigma}{1 + S \rho_p / \sigma}} \quad (32)$$

where

$$\sigma = A \frac{\mu}{d^2} \left(\frac{\rho_p}{\rho_f} \right)^n \quad (33)$$

A is a constant which Bradbury estimates to be about 27, $\mu = 1.81 \times 10^{-5}$, d is the diameter of the fibers (28 microns for wool), ρ_f is the absolute density of the fibers, ρ_p is packing density, traditionally 8 kg/m^3 (0.5 lb/cu. ft.). It should be noted that Bradbury's paper contains a misprint and eq.(32) is the correct form.

Eq.(23) may be re-written as

$$Z'_L = \frac{\rho_0 c}{s_L} \gamma \quad (34)$$

and the new wavenumber, k' , may be written as

$$k' = k\gamma \quad (35)$$

The only thing that remains is to define the velocity at the line exit. This is fairly straightforward and is obtained by applying eq. (17),(20), and (21):

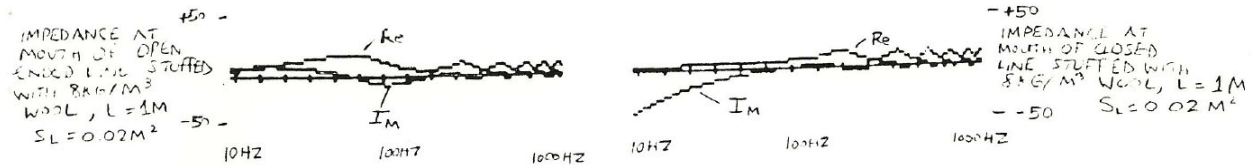
$$u_v = u_D e^{-jkL} \left[\frac{2}{1 + \frac{Z_v}{Z_L}} \right] \quad (36)$$

where Z_v is taken from eq.(25) and Z_L from eq.(23) or (35). This equation also accounts for the fact that if the line is closed, there will be no output at the exit.

The mechanical impedance seen by the driver changes radically when the line is filled, and as packing density increases, the variations in impedance are smoothed. Also, observe that while the unstuffed line presents a highly reactive load to the driver, the stuffed line impedance is substantially resistive.

The total acoustic pressure at a distance r is

$$\frac{-1}{2\pi r} \rho_0 (u_D S_D - u_v S_v) j s \quad (37)$$

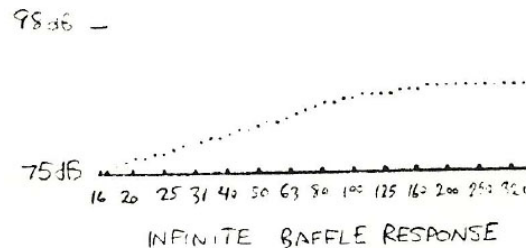


It should be recognized that the other enclosures--closed, vented, or horn loaded--present an essentially reactive load to the driver. The TL is unique in that its primary function is to dissipate energy.

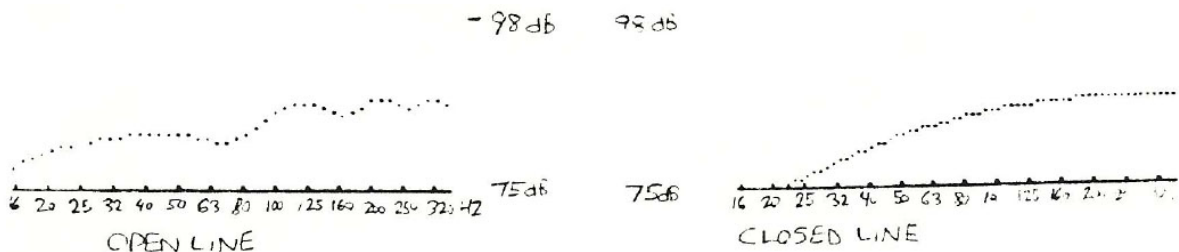
DESIGN EXAMPLES

It is now possible to examine the behavior of a system which includes a driver. For the first example, I have selected a unit which has traditionally been favored by TL enthusiasts, the Audax 10" woofer HD24B45. It has the following specifications:

Re	6 ohms
fs	23 Hz
Cms	.0014 m/N
Qms	4.05
Qes	0.33
Qts	0.31
Rms	1.15 kg/s
Mmd	.0337 kg
Sd	.037 m ²
Bl	11.4 N/A
Sensitivity	90dB SPL 2.83V/1m

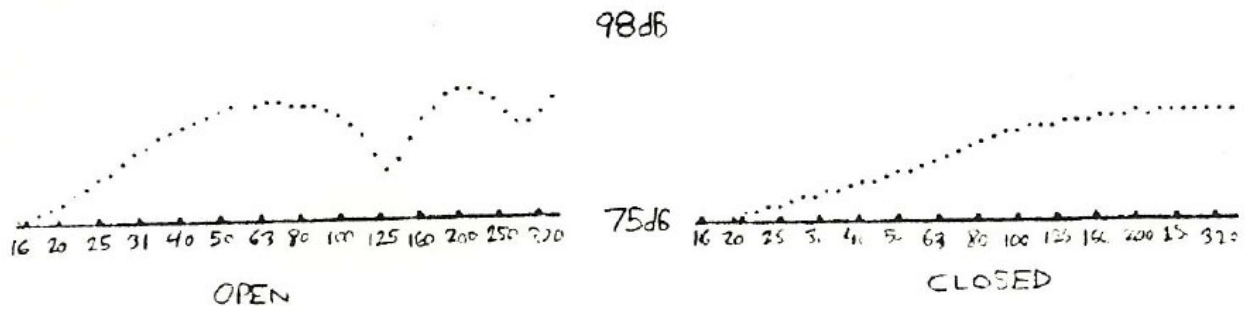


Based on construction plans in Speaker Builder magazine [5], the length of the line is 3.5m and cross section is .059m². Packing density is 8kg/m³.

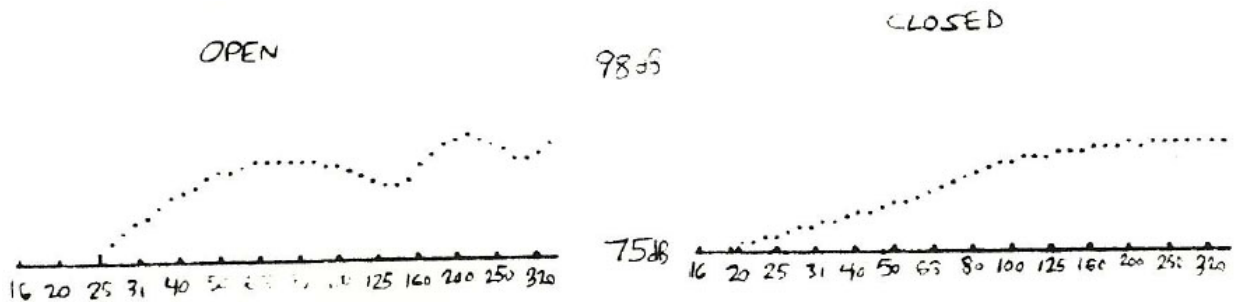


While the open line curve may appear to have excessive variations, it has several desirable characteristics. First, its output is strong at very low frequencies. Second, its gentle sloping characteristic indicates absence of pronounced resonances and rapid phase shifts. Third, this curve is roughly the inverse of the response of a speaker near typical room boundaries, as Allison [1] has shown.

If the length of the line is reduced from 3.5 to 2 meters, the response curve below results. When closed, this line offers more apparent bass extension than the 3.5 meter line (this is due to the location of the first resonance at about 25 Hz instead of 13 Hz). The open line is clearly inferior, exhibiting large cancellations due to the output at the end of the pipe.



If the cross section of the line is reduced from 0.059m^2 to 0.037m^2 (the area of the cone) it is clear that the additional resistance caused by the constriction reduces the low bass response.

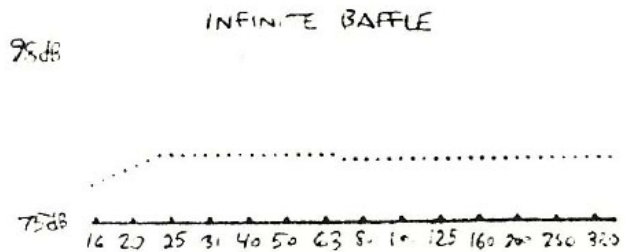


The response of the system is always limited by the inherent characteristics of the driver. If greater bass extension is desired, then a driver with greater sensitivity at low frequencies is needed.

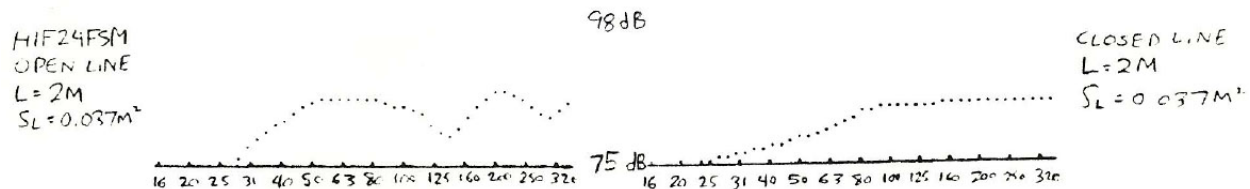
At resonance, a driver in an infinite baffle with a specified Q_t will have a sensitivity of $20 \cdot \log(Q_t)$ dB relative to the mid band. Therefore we will expect a driver with a higher Q to be more responsive at low frequencies. The above woofer's response is down 10 dB at resonance.

With this in mind, I decided to try a new woofer, an Audax HIF24FSM with the following specifications:

R_e 6.5 ohms
 f_s 29 Hz
 C_{ms} .00156 m/N
 Q_{ms} 3.81
 Q_{es} 0.80
 Q_{ts} 0.66
 R_{ms} 0.97 kg/s
 M_{md} .0195 kg
 S_d .0301 m²
 Bl 5.4 N/A
 Sensitivity 86dB SPL 2.83v/lm

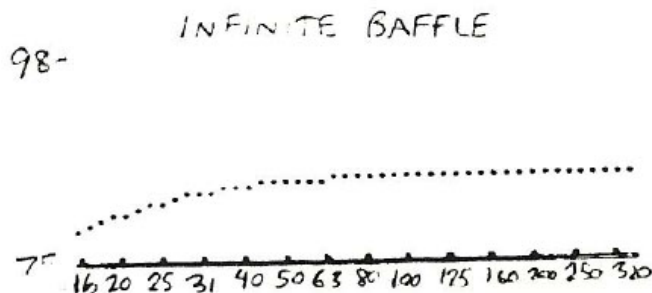


This driver is expected to perform better because at resonance it is only down 3.5 dB. However, when we put this driver in the same enclosure as before, both the open and closed lines cause considerable reductions in low frequency output. In fact, this woofer has even less output at low frequencies than the first one did. The reason for this is that the HIF24FSM has a low cone mass and small force factor Bl . It is therefore more sensitive to external effects because of its low mechanical impedance. It stands to reason that a driver with a high mechanical impedance (higher M_{md} , higher Bl) will be less susceptible to this problem.

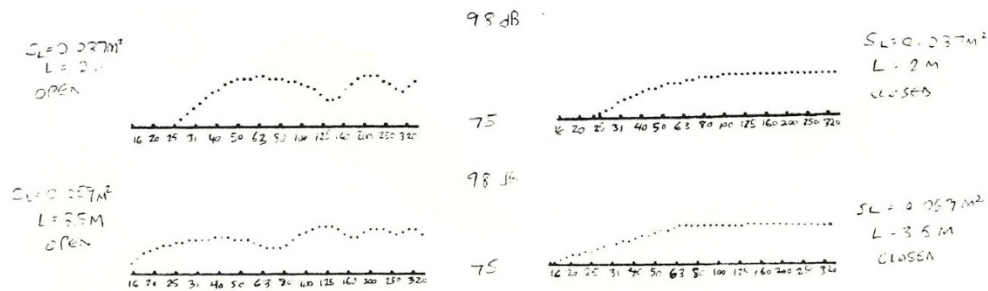


Following this idea, I will model another woofer, the Madisound 10208, with the following specifications:

R_e 5.5 ohms
 f_s 22 Hz
 C_{ms} .00106 m/N
 Q_{ms} 2.9
 Q_{es} 0.56
 Q_{ts} 0.47
 R_m 2.37
 M_{md} 0.050 kg
 S_d .035 m²
 Bl 8.2 N/A
 Sensitivity 87 dB 2.83v/lm



Its curves for both open and closed boxes (in the same 2m enclosure) are virtually identical to the HD24B45 curves below 100 Hz and simply show lower sensitivity above that point.



This is also true when it is put back into the full sized, 3.5m enclosure; both woofers have identical bass extension but with sensitivity differences in the mid band. It is evident that a transmission line can largely determine the response characteristic of any driver at low frequencies.

An interesting feature of the closed TL is that making the box too small (within reason) causes a gradual roll-off at low frequencies; any other enclosure made too small would have a peak in the mid bass and sharp rolloff below. Once again, this is because of the reactive nature of the other enclosures.

THE ACCURACY OF THE MODEL

Unfortunately, I am unable to provide measurements to support the theory which I have developed. However, the results are fully in keeping with my expectations, and discrepancies can plausibly be accounted for by the following:

1. The model used in this paper assumes a constant cross section, while real enclosures are almost always folded and tapered. Tapering conserves space, has less effect on the driver than making the entire line smaller, and reduces the output at the line exit and hence the cancellation effects at mid frequencies.
2. The approach taken in this paper can be extended in a fairly straightforward manner simply by modeling the enclosure as not one single transmission line, but several cascaded together.
3. A simpler way to deal with this might be to model the enclosure as an unfolded tapered pipe, as described by Olson [11].

CONCLUSIONS

Time and space limit full exploration of the many variations possible with this design. However, several observations and recommendations can be made:

1. The TL is not a "free lunch" any more than the other enclosures are; its restrictions on cabinet size are similar to those of other enclosures. It simply represents a different set of compromises.
2. It has the advantage of a gradual rolloff at low frequencies, which can be used advantageously in combination with room boundaries.
3. The cross-section of the line should exceed the area of the woofer cone in the first half of the line.
4. The closed TL is a viable alternative to the open line, and unlike the acoustic suspension enclosure, strongly damps the fundamental resonance of the driver.
5. The response of a closed TL at extremely low frequencies is identical to that of an ordinary closed box, reducing infrasonic cone motion.
6. The length of an open TL primarily affects the low frequency cancellation caused by the vent. Since these materials reduce the speed of sound by a factor of two or so, the length of the line can be similarly reduced. The rule of thumb that the TL should be $1/4 \lambda$ long is reasonable as long as λ is adjusted for the altered speed of sound at low frequencies. The quarter-wavelength, anti-resonance characteristic of the open line is nullified by the fibrous material in the line.

I believe that the TL is favored by many audiophiles because when properly constructed, it is a virtually non-resonant enclosure, presenting to the driver an essentially resistive load. It is also somewhat idiot proof; that is to say, as long as the designer follows some basic guidelines, it can't sound too bad; a horn, on the other hand, is a disaster without careful attention (many that have had careful attention are still disasters, in my opinion). Perhaps that is why the TL has stood the test of time without a theory to back it up.

FURTHER WORK

The first thing that needs to be done is to use a computer to model many different variations of what I've already done in the few examples included here. My calculator is just too slow for that.

Secondly, Bradbury acknowledges in his paper that the characteristics of fibrous materials have not been fully investigated; work remains to be done, both theoretical and experimental.

Finally, I'm anxious to test my theory in a lab using a real Transmission Line.

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